

No. of Functional Features Included in the Solution

Date	10 JULY 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	USER TEXT INPUT	Accepts user text for emotion analysis	Frontend (Flask UI)	DONE	HIGH
2	Text Preprocessing	Cleans and tokenizes user input before prediction	Preprocessing Module	DONE	HIGH
3	Emotion Detection	Predicts one of seven emotions using DistilBERT and BiLSTM models	ML Prediction Engine	DONE	HIGH
4	Hybrid Prediction	Combines multiple models for improved prediction accuracy	Hybrid Classifier	DONE	HIGH
5	AI Learning Support	Generates personalized learning suggestions using Google Gemini API	Recommendation Module	DONE	HIGH
6	Prediction History	Saves prediction results with timestamps for future reference	Hybrid Clas	DONE	MEDIUM
7	Interactive Web Interface	Provides an easy-to-use interface for users	Flask Frontend	DONE	MEDIUM
8	Error Handling & Validation	Validates input and handles invalid or empty submissions gracefully	Backend	DONE	MEDIUM

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Text input form, Prediction display
2	Backend / Logic	4	Preprocessing, Emotion Prediction, Hybrid Model, Error Handling
3	Database / Storage	1	Prediction history stored in CSV files
4	API / Integration	1	Google Gemini API for personalized learning recommendations
5	Security / Authentication	0	Authentication planned for future versions